

Floorplanning, Macro Placement & Pin Placement

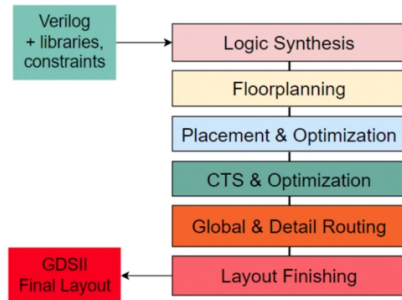
Paula Heinz

Research Institute for Discrete Mathematics

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Outline

- ① **Floorplanning:** canvas, rows, routing grid
- ② **Macro placement:** large, rigid rectangles
- ③ **Pin placement:** legal boundary sites for I/O pins
- **Cell:** shape, pins, timing/power models
- **Macro:** large hard block (e.g. SRAM, PLL, hardened IP)
- **Pin:** external connection / legal metal shape



Three Coupled Optimization Problems

Floorplanning

Choose:

- die/core geometry
- row structure
- routing tracks
- macro/pin constraints

Pin placement

For pins p_i :

$$p_i \in S_{\text{legal}}$$

minimize interconnect cost s.t.
layers, spacing, exclusions,
grouping are respected

Macro placement

For rectangles M_i :

$$(x_i, y_i, o_i)$$

subject to

$$M_i \cap M_j = \emptyset$$

and outline/halo/channel
constraints.

Why not one giant optimization?

Heterogeneous search space: continuous coordinates, discrete orientations/slots, topology, hard geometric constraints $\Rightarrow$ solve structured subproblems sequentially.

Floorplan Initialization

`initialize_floorplan`: automatic (utilization / aspect ratio) or explicit coordinates.

```
initialize_floorplan \  
  -utilization 70 \  
  -aspect_ratio 1.0 \  
  -core_space 10 \  
  -site FreePDK45_38x28_10R_NP_162NW_340
```

```
initialize_floorplan \  
  -die_area {0 0 2000 2000} \  
  -core_area {100 100 1900 1900} \  
  -site FreePDK45_38x28_10R_NP_162NW_340
```

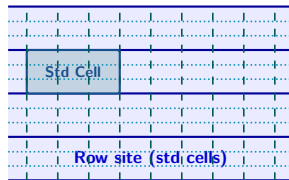
`initialize_floorplan` resets routing tracks; `make_tracks` creates routing tracks.

Key terms

die: full silicon rectangle incl. I/O ring

core: inner area for cells & macros (inset: *core space*)

site: template for standard-cell placement rows



— Placement rows ■ Standard cell
... Horiz. tracks -- Vert. tracks

Pin Placement

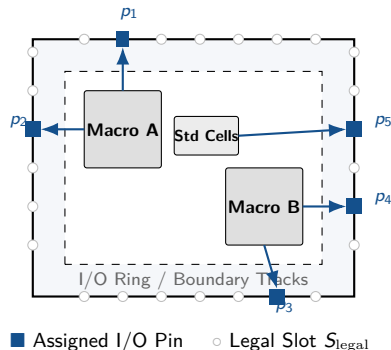
Pin placement

For I/O pins p_i :

$$p_i \in S_{\text{legal}}$$

minimize interconnect cost s.t. layers, spacing, exclusions, grouping are respected.

- Assigns top-level I/O pins to legal boundary slots / routing tracks along the die perimeter.
- Connects boundary pins to internal standard cells and macros while minimizing wirelength.



Pin Placement as Assignment Problem

Pins $P = \{1, \dots, m\}$, legal boundary slots $S = \{1, \dots, n\}$ ($n \geq m$).

$$\begin{aligned} \min_x \quad & \sum_{i \in P} \sum_{j \in S} c_{ij} x_{ij} \\ \text{s.t.} \quad & \sum_{j \in S} x_{ij} = 1 \quad \forall i \in P \\ & \sum_{i \in P} x_{ij} \leq 1 \quad \forall j \in S \\ & x_{ij} \in \{0, 1\} \quad \forall i \in P, j \in S \end{aligned}$$

- c_{ij} : estimated wirelength cost (pin i at slot j)
- Totally unimodular constraint matrix $\Rightarrow$ integral LP relaxation
- Solvers: **Hungarian algorithm** ($O(m^2n)$ or $O(n^3)$ where $n \geq m$, exact) / **Simulated Annealing** (-annealing, default)

The Hungarian Algorithm

For m pins and n slots ($m \leq n$), augment cost matrix $C \in \mathbb{R}^{m \times n}$ to square $n \times n$ with $n - m$ **dummy pins** (cost 0):

Algorithm steps ($n \times n$):

- 1 **Row reduction**: subtract each row minimum
- 2 **Column reduction**: subtract each column minimum
- 3 **Zero covering**: cover all zeros with the minimum number of lines
- 4 **Matrix adjustment** ($< n$ lines): subtract the smallest uncovered value and add it at line intersections $\rightarrow$ step 3
- 5 **Assignment** (n lines): select independent zeros; discard dummy assignments

Example (3×3)

$$\begin{pmatrix} 4 & 2 & 8 \\ 2 & 3 & 7 \\ 3 & 1 & 6 \end{pmatrix} \xrightarrow{\text{row}} \begin{pmatrix} 2 & 0 & 6 \\ 0 & 1 & 5 \\ 2 & 0 & 5 \end{pmatrix} \xrightarrow{\text{col}} \begin{pmatrix} 2 & 0 & 1 \\ 0 & 1 & 0 \\ 2 & 0 & 0 \end{pmatrix}$$

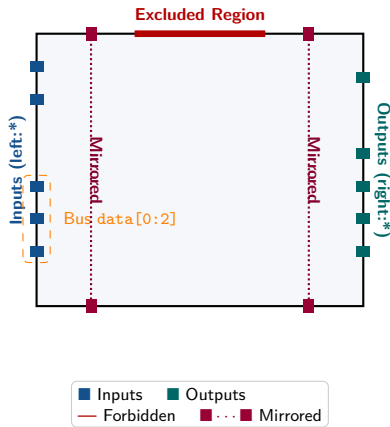
Assignment from zeros:

$$1 \rightarrow 2, 2 \rightarrow 1, 3 \rightarrow 3 \quad (\text{cost } 2 + 2 + 6 = 10)$$

Complexity: $O(n^3)$, exact minimum-cost matching

Physical Constraints on Pins

- **Grid / Track:** valid routing tracks & metal layers
- **Forbidden regions:** excluded edge intervals (`exclude_io_pin_region`)
- **Directional edges:** restrict inputs/outputs to specific edges (`set_io_pin_constraint -direction`)
- **Pin grouping / buses:** keep bus bits ordered & adjacent (`-group`)
- **Mirrored pin pairs:** aligned coordinates across opposite edges for block abutment (`-mirrored`)

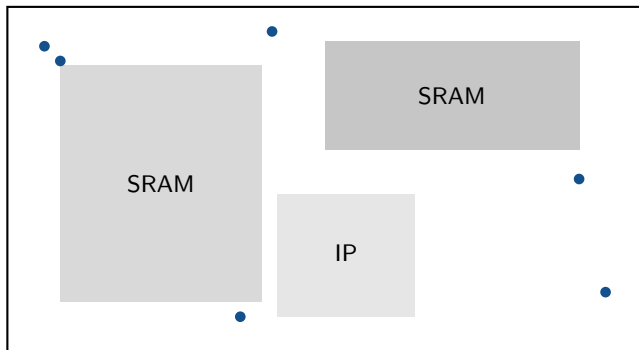


Pin Placement: OpenROAD Commands

```
set_io_pin_constraint \  
-direction input \  
-region left:*  
  
set_io_pin_constraint \  
-direction output \  
-region right:*  
  
place_pins \  
-hor_layers metal3 \  
-ver_layers metal2 \  
-min_distance_in_tracks \  
-annealing \  
-write_pin_placement results/pins.tcl
```

- -hor_layers: metal layers for top/bottom edge pins
- -ver_layers: metal layers for left/right edge pins
- -annealing: simulated annealing engine (vs. Hungarian)
- -write_pin_placement: reproducible place_pin script output

Macro Placement

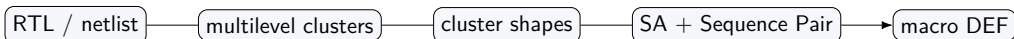


Macro geometry: corridors, blockages, long-range constraints

- Critical impact on downstream cell placement and routing
- Fixed dimensions; degrees of freedom: (x, y) location and orientation
- Problem formulation: **constrained rectangle packing** + optimization of wirelength, space utilization, congestion, ...

Hier-RTLMP: Hierarchical Macro Placement

- ① **Autoclustering**: multilevel physical hierarchy from RTL hierarchy + dataflow
- ② **Cluster shaping/placement**: feasible footprints for macro/std-cell clusters
- ③ **Leaf cluster macro placement**: Sequence Pair representation + simulated annealing
- ④ **Multi-objective cost**: area, HPWL, outline, fence/guidance, notch/blockage penalties



Strategy

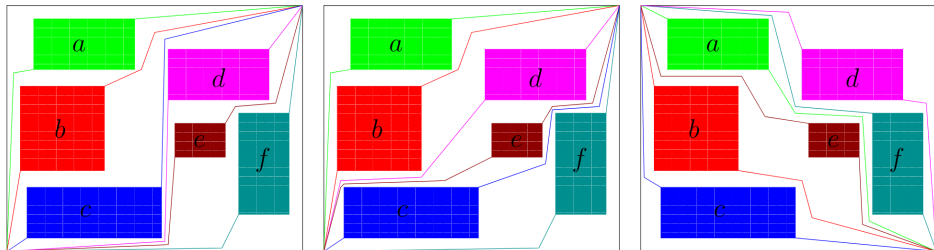
Decomposition of flat packing into a sequence of smaller subproblems; hierarchy- and dataflow-aware.

Sequence Pair Representation

Encoding: (π^+, π^-) represents relative 2D positions:

	1	2	3	4	5	6
π^+	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>
π^+	<i>a</i>	<i>b</i>	<i>d</i>	<i>e</i>	<i>c</i>	<i>f</i>
π^-	<i>c</i>	<i>b</i>	<i>e</i>	<i>a</i>	<i>f</i>	<i>d</i>

π^+ / π^- can be constructed non-uniquely from a placement using **step-lines**



Geometric Relations ($i \neq j$):

- i before j in π^+ and $\pi^- \implies i$ is **left** of j
- i before j in π^+ , i behind j in $\pi^- \implies i$ is **below** j

Coordinate Decoding:

- Build horizontal (G_H) and vertical (G_V) constraint directed acyclic graphs.
- Longest path in DAG gives (x, y) .

Simulated Annealing for Macro Placement

Neighborhood Moves

Perturb sequence pair (π^+, π^-) :

- **M1:** Swap two elements in π^+ or π^-
- **M2:** Swap two elements in both π^+ and π^-
- **M3:** Rotate macro orientation
- **M4:** Resize soft cluster footprint

Acceptance & Cooling

Cost delta $\Delta C = C_{\text{new}} - C_{\text{old}}$:

- If $\Delta C \leq 0$: always accept move
- If $\Delta C > 0$: accept with probability

$$P(\text{accept}) = \exp\left(-\frac{\Delta C}{T}\right)$$

Cooling schedule: $T_{k+1} = \alpha T_k$ ($\alpha \approx 0.95$).
Early exploration of floorplan topologies $\rightarrow$ late convergence to good packing.

Macro Placement: OpenROAD Commands

```
set_macro_base_halo 10 10

rtl_macro_placer \  
    -report_directory reports/macro \  
    -write_macro_placement results/macro_place.tcl

place_macro \  
    -macro_name u_sram0 \  
    -location {420 760} \  
    -orientation R0
```

- `set_macro_base_halo`: halo / keepout margin around macros
- `rtl_macro_placer`: hierarchical macro placer invocation
- `-write_macro_placement`: export generated `place_macro` Tcl script

Hands-on Exploration

- Baseline run → parameter variations (`-target_util`, `-wirelength_weight`, guidance/fence constraints) → solution comparison.
- Manual placement of critical SRAM → automatic placement of remainder → compare wirelength and congestion.

Takeaways

- ① **Floorplanning**: geometric canvas, core/die dimensions, rows, track grids
- ② **Pin placement**: boundary assignment; Hungarian matching ($O(m^2n)$ / $O(n^3)$) or simulated annealing
- ③ **Macro placement**: constrained rectangle packing; **Hier-RTLMP** (multilevel clustering + Sequence Pair + simulated annealing)

References



S. Held, J. Vygen, *Chip Design*, Lecture Notes. Preliminary version: July 11, 2023.



OpenROAD Documentation, *Initialize Floorplan*.

<https://openroad.readthedocs.io/en/latest/main/src/ifp/README.html>



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A. B. Kahng, R. Varadarajan, Z. Wang, *Hier-RTLMP: A Hierarchical Automatic Macro Placer for Large-scale Complex IP Blocks*, 2023.